ALL τ_k-MAXIMAL GRAPHS HAVE THE SAME SIZE

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ABSTRACT. A graph is τ_k-maximal if none of its subgraphs has $k+1$ edge-disjoint spanning trees, but adding any missing edge creates such a subgraph. Wang and Tian conjectured that every τ_k-maximal graph on $n \geq 2k+2$ vertices has exactly $(k+1)(n-1)-1$ edges. We prove this conjecture.

1. INTRODUCTION

All graphs in this note are finite and simple. For a graph H with at least two vertices, $\tau(H)$ denotes the largest number of edge-disjoint spanning trees of H, and for a graph G we put

$$\bar{\tau}(G) = \max\{\tau(H) : H \subseteq G,\ |V(H)| \geq 2\}.$$

Adding an edge raises $\bar{\tau}$ by at most one. Following Wang and Tian [6], a graph G is τ_k-*maximal* if $\bar{\tau}(G) \leq k$ while $\bar{\tau}(G+e) \geq k+1$ for every edge e of the complement of G. Since $\tau(K_n) = \lfloor n/2 \rfloor$, the complete graph K_n is the only τ_k-maximal graph on $n \leq 2k+1$ vertices, so the interesting range is $n \geq 2k+2$. There Wang and Tian proved that a τ_k-maximal graph has at most $(k+1)(n-1)-1$ edges, built a family of τ_k-maximal graphs with exactly that many edges, and proved that every τ_1-maximal graph on n vertices has $2n-3$ edges. They proposed the following statement [6, Conjecture 1].

Conjecture 1. Let $k \geq 1$ and $n \geq 2k+2$. Every τ_k-maximal graph of order n has exactly $(k+1)(n-1)-1$ edges.

We prove it, together with a description of the τ_k-maximal graphs by counting alone. For $X \subseteq V(G)$ let $e_G(X)$ be the number of edges of G with both ends in X.

Theorem 1.1. *Let $k \geq 1$ and $n \geq 2k+2$, and let G be a graph with n vertices. The following are equivalent.*

(a) *G is τ_k-maximal.*
(b) *$|E(G)| = (k+1)(n-1)-1$, and $e_G(X) \leq (k+1)(|X|-1)-1$ for every $X \subseteq V(G)$ with $|X| \geq 2$.*

In particular, Conjecture 1 is true.

The proof has three steps. Section 2 turns the condition $\bar{\tau}(G) \leq k$ into a counting condition, using the theorem of Nash-Williams and Tutte. Section 3 shows, by an exchange argument with tight sets, that all maximal edge sets satisfying the counting condition have the same size; in the language of matroids they are the bases of a count matroid. Section 4 computes the common size with one explicit example.

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2. The counting condition

Throughout, $a = k+1 \geq 2$. Call a graph G *a-sparse* if

$$e_G(X) \leq a(|X|-1)-1 \qquad \text{for every } X \subseteq V(G) \text{ with } |X| \geq 2. \tag{2.1}$$

We use the theorem of Nash-Williams [4] and Tutte [5]: a graph H has a edge-disjoint spanning trees if and only if, for every partition $\mathcal{P}$ of $V(H)$ into t parts, at least $a(t-1)$ edges of H join different parts.

Lemma 2.1. *For every graph G, $\bar{\tau}(G) \leq k$ if and only if G is $(k+1)$-sparse.*

Proof. If a subgraph H of G on a vertex set X with $|X| \geq 2$ has a edge-disjoint spanning trees, then $e_G(X) \geq |E(H)| \geq a(|X|-1)$, and (2.1) fails for X.

Conversely, suppose (2.1) fails, and choose a set X with $|X| \geq 2$ and $e_G(X) \geq a(|X|-1)$ with $|X|$ as small as possible. Let $\mathcal{P} = \{P_1, \ldots, P_t\}$ be a partition of X with $t \geq 2$. A part with at least two vertices is a smaller set than X, so by the choice of X it spans at most $a(|P_i|-1)-1 < a(|P_i|-1)$ edges; a part with one vertex spans $0 = a(|P_i|-1)$ edges. Hence the number of edges of $G[X]$ joining different parts is at least

$$e_G(X) - \sum_{i=1}^{t} a(|P_i|-1) \;\geq\; a(|X|-1) - a(|X|-t) = a(t-1).$$

By the theorem of Nash-Williams and Tutte, $G[X]$ has a edge-disjoint spanning trees, so $\bar{\tau}(G) \geq a = k+1$. □

By Lemma 2.1, a graph G on the vertex set V of K_n is τ_k-maximal exactly when $E(G)$ is $(k+1)$-sparse and $E(G) \cup \{e\}$ is not, for every edge e of K_n outside $E(G)$. In other words, $E(G)$ is a maximal sparse subset of $E(K_n)$; here and below we call a set F of edges sparse when the graph (V, F) is a-sparse.

3. Tight sets

Fix a sparse set I of edges of K_n, and write $e(X) = e_I(X)$. A set $X \subseteq V$ with $|X| \geq 2$ is *tight* if $e(X) = a(|X|-1)-1$, that is, if (2.1) holds for X with equality.

Lemma 3.1. (i) *$e(X \cup Y) + e(X \cap Y) \geq e(X) + e(Y)$ for all $X, Y \subseteq V$.*

(ii) *If X and Y are tight and $|X \cap Y| \geq 2$, then $X \cup Y$ is tight.*

(iii) *If uv is an edge of K_n outside I and $I \cup \{uv\}$ is not sparse, then some tight set contains both u and v.*

Proof. (i) An edge counted in $e(X)$ or in $e(Y)$ lies inside $X \cup Y$, and an edge counted in both lies inside $X \cap Y$.

(ii) Since $|X \cap Y| \geq 2$, the set $X \cap Y$ satisfies (2.1), and by (i)

$$e(X \cup Y) \;\geq\; \big(a(|X|-1)-1\big) + \big(a(|Y|-1)-1\big) - \big(a(|X \cap Y|-1)-1\big) = a(|X \cup Y|-1)-1.$$

The reverse inequality is (2.1) for $X \cup Y$.

(iii) Some X with $|X| \geq 2$ has more than $a(|X|-1)-1$ edges of $I \cup \{uv\}$ inside it, and at most $a(|X|-1)-1$ edges of I. So uv lies inside X, and X is tight. □

Lemma 3.2. *Let I be a maximal sparse subset of $E(K_n)$ and let J be any sparse subset. Then $|J| \leq |I|$.*

Proof. Tight sets refer to I. Let $M_1, \dots, M_p$ be the tight sets that are maximal under inclusion; every tight set lies inside one of them. If $|M_i \cap M_j| \geq 2$, then $M_i \cup M_j$ is tight and contains both, so $M_i = M_i \cup M_j = M_j$. Hence two different sets M_i share at most one vertex, and every edge of K_n lies inside at most one of them. Moreover, if $uv \notin I$, then $I \cup \{uv\}$ is not sparse because I is maximal, so by Lemma 3.1(iii) the pair uv lies inside some M_i.

Let R be the set of edges of K_n lying inside no M_i; by what we just proved, $R \subseteq I$. Counting the edges of J according to the set M_i they lie in,

$$|J| = \sum_{i=1}^{p} e_J(M_i) + |J \cap R| \;\leq\; \sum_{i=1}^{p} \big(a(|M_i| - 1) - 1\big) + |R| = \sum_{i=1}^{p} e_I(M_i) + |I \cap R| = |I|.$$

Here $e_J(M_i) \leq a(|M_i| - 1) - 1$ because J is sparse, and $a(|M_i| - 1) - 1 = e_I(M_i)$ because M_i is tight. □

Applying Lemma 3.2 in both directions gives the following.

Corollary 3.3. *All maximal sparse subsets of $E(K_n)$ have the same size r, the largest size of a sparse subset. A graph on n vertices is τ_k-maximal if and only if it is $(k+1)$-sparse and has exactly r edges.*

Remark 3.4. In the language of matroids, Lemma 3.2 says that the sparse sets are the independent sets of a matroid on $E(K_n)$. This is a special case of a theorem of Edmonds [1] on matroids defined by submodular functions, and such count matroids are well known in rigidity theory [3, 7]. The direct argument above keeps the note self-contained.

4. The rank

Taking $X = V$ in (2.1) shows $r \leq a(n-1) - 1$. For the reverse inequality we need one sparse graph with $a(n-1) - 1$ edges; Wang and Tian give a family of them [6], and the following one is quick to check.

Lemma 4.1. *Let $n \geq 2a$. Split the vertex set into A with $|A| = a$ and B with $|B| = n - a \geq a$, fix $W \subseteq B$ with $|W| = a$, and fix two vertices w, w' of W. Let G_0 have as edges all pairs inside A, all pairs between A and B, and all pairs inside W except ww' (Figure 1). Then G_0 is a-sparse and has $a(n-1) - 1$ edges.*

Proof. The number of edges is $\binom{a}{2} + a(n-a) + \binom{a}{2} - 1 = a(n-1) - 1$.

Put $C = A \cup W$. On C the graph G_0 is the complete graph on $2a$ vertices without the edge ww', and each vertex outside C is adjacent exactly to the vertices of A. We first check (2.1) for sets $Y \subseteq C$ with $y = |Y| \geq 2$. If $y = 2a$, then $Y = C$ and $e_{G_0}(Y) = \binom{2a}{2} - 1 = a(2a-1) - 1$, which is (2.1) with equality. If $y \leq 2a - 1$, then $(y-1)(2a-y) \geq 2$: for $y = 2$ the product is $2a - 2 \geq 2$, and for $y \geq 3$ the first factor is at least 2 and the second at least 1. This rearranges to $y(y-1) \leq 2a(y-1) - 2$, so

$$e_{G_0}(Y) \leq \binom{y}{2} \leq a(y-1) - 1.$$

Now let $X \subseteq V$ with $|X| \geq 2$, and put $Y = X \cap C$ and $Z = X \setminus C$. There are no edges inside Z, and a vertex of Z has at most $|X \cap A| \leq \min(a, |Y|)$ neighbours in X. Hence

$$e_{G_0}(X) \leq e_{G_0}(Y) + |Z| \min(a, |Y|).$$

If $|Y| \geq 2$, this is at most $a(|Y| - 1) - 1 + a|Z| = a(|X| - 1) - 1$. If $|Y| = 1$, it is at most $|Z| \leq a|Z| - 1 = a(|X| - 1) - 1$, as $|Z| \geq 1$ and $a \geq 2$. If $Y = \emptyset$, it is $0 \leq a(|X| - 1) - 1$. □

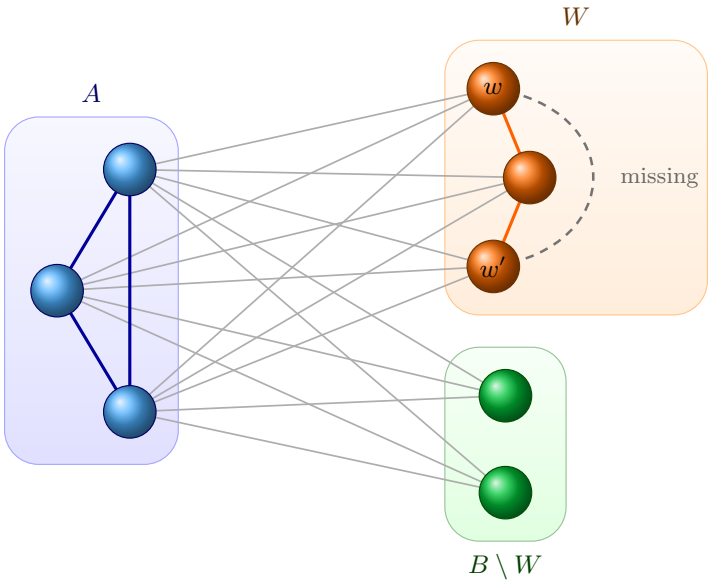


FIGURE 1. The graph G_0 for $a = 3$ and $n = 8$. On $A \cup W$ it is the complete graph K_6 without the edge ww', and each vertex of $B \setminus W$ is joined to the three vertices of A. It has $20 = 3 \cdot 7 - 1$ edges.

Proof of Theorem 1.1. By Lemma 4.1 and the upper bound above, $r = a(n-1) - 1 = (k+1)(n-1) - 1$ when $n \geq 2k+2$. Corollary 3.3 now says that G is τ_k-maximal exactly when it is $(k+1)$-sparse and has $(k+1)(n-1) - 1$ edges, which is condition (b). □

Remark 4.2. Lemma 2.1 holds for every n, and for $n \leq 2k+1$ the whole edge set of K_n is $(k+1)$-sparse, since a set X with $2 \leq |X| \leq 2k+1$ spans $\binom{|X|}{2} \leq (k+1)(|X|-1) - 1$ edges. So in that range the rank is $\binom{n}{2}$ and K_n is the only τ_k-maximal graph, in line with the observation of Wang and Tian. The proof above also gives back their upper bound and their result for $k = 1$.

Remark 4.3. For $k = 1$, condition (b) says $|E(G)| = 2n - 3$ and $e_G(X) \leq 2|X| - 3$ whenever $|X| \geq 2$. These are the Laman graphs, the generically minimally rigid graphs in the plane [2]. So the τ_1-maximal graphs are exactly the Laman graphs, and in general the τ_k-maximal graphs are the bases of the $(k+1, k+2)$ count matroid.

Remark 4.4. Theorem 1.1 also says that a τ_k-maximal graph can be grown greedily: starting from any $(k+1)$-sparse graph on $n \geq 2k+2$ vertices and adding edges one at a time as long as the graph stays sparse always ends with $(k+1)(n-1) - 1$ edges. Whether a given set of edges is sparse can be decided in polynomial time, for instance with the pebble game of Lee and Streinu [3].

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Data availability. The proofs in this article are complete by hand. The source of the article and the programs that check its statements (exhaustive checks in C and Python on every graph with at most seven vertices, further checks in C++, Bash and Julia, and a Lean 4 proof with Mathlib of Lemmas 3.1, 3.2 and 4.1 and of Theorem 1.1 in sparse form for every n and k) are in release v1.1.0 of the repository https://github.com/creelie/eConjectureTauMaximal, https://github.com/creelie/eConjectureTauMaximal/releases/tag/v1.1.0.

Use of generative AI. During the preparation of this work the author used Claude (Anthropic) to search the literature, to find and check the proofs, and to draft the text. The author checked every step of the proofs and every reference, and takes full responsibility for the content.

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